Rituraj Singh

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Summary

Final-year CSE (AI & ML) engineer building reliable, automated systems for industry: production AI at an operating plant, manufacturing IoT pipelines and CI/CD-gated releases.

Education

VIT Bhopal University 2023 – 2027
B.Tech, Computer Science and Engineering (AI & ML); CGPA 8.39 / 10 *Bhopal, MP*
Leadership: Co-Lead, Technical Team, Hindi Club (2025 – 2026)

Experience

ConsultiFI Data Jun 2026 – Present
Software Developer Intern *Remote*

- Automated anomaly detection on continuous IoT flow-meter data from manufacturing clients using unit-tested Python pipelines, replacing manual review of readings.
- Shipped every release through CI gated on Playwright E2E tests, backups and health checks.
- Core team member on a multi-tenant SaaS platform for 200 client organisations and 10,000 concurrent users, with strict per-client data isolation.
- Took the first client live without a system rewrite; integrated WhatsApp chatbots.

Tata Steel Utilities and Infrastructure Services Ltd May 2025 – Jun 2025
AI Intern *Jamshedpur, JH*

- Built and deployed real-time vehicle detection and alerting on 4 live CCTV feeds at an operating plant, replacing manual monitoring.
- 92% production accuracy on 1,000+ daily detections; cut manual monitoring effort by ∼60%.
- Owned the model from data preparation to evaluation; presented results to plant stakeholders and wrote the handover documentation.

Projects

Praxya: LLM Document Extraction for Regulatory Reporting Mar 2026 – Present
Python, FastAPI, PostgreSQL, Pydantic, LLM APIs, OCR

- Built an LLM pipeline that turns unstructured plant bills and invoices into audit-ready Scope 1 & 2 emissions reports for SEBI BRSR filings.
- A fault-tolerant background worker runs OCR then LLM extraction with up to 3 retries; Pydantic schemas validate every extracted figure.
- Human sign-off on every figure; insert-only lineage tables trace each value to its source.
- Exposed FastAPI REST endpoints for document upload, job-status tracking and report export, with asynchronous processing so large uploads never block the API.
- Mapped extracted fuel and electricity quantities to standard emission factors to compute Scope 1 and Scope 2 totals per facility.

FileMyCa: GST Data-Collection Portal for CA Firms Feb 2026 – Mar 2026
Next.js, TypeScript, Supabase (PostgreSQL)

- Shipped solo in 5 days: a document-upload portal and GST filing dashboard for CA firms.
- PostgreSQL row-level security restricts each firm to its own clients' financial data.
- Role-based access with Supabase Auth gives CA staff a firm-wide dashboard and clients a separate upload view.
- Dashboard tracks pending documents and filing status per client for each GST period, so firms see what is missing before deadlines.

Skills

Programming: Python, C++, SQL, TypeScript, FastAPI, Django, PostgreSQL, REST APIs
Cloud & DevOps: AWS, Docker, Git, GitHub Actions (CI/CD), Playwright, unit testing
Applied AI: LLM integration, RAG, LangChain, LangGraph, OCR, OpenCV, Claude Code
Fundamentals: DSA, OOP, DBMS, Operating Systems, Computer Networks